Volumes by the Disc and Washer Methods

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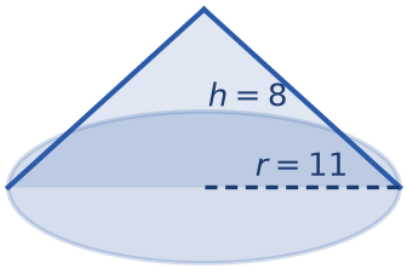
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The Disc Method Around Coordinate Axes**

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 11]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

2 **[Foundational]** • **Target: The Washer Method Around Coordinate Axes**

The base of a solid is the region under $y = 1\sqrt{x}$ on $[0, 11]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.



3 **[Foundational]** • **Target: Revolutions Around Non-Axis Horizontal Lines**

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 7]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

4 **[Foundational]** • **Target: Volumes with Known Cross-Sections**

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 9]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

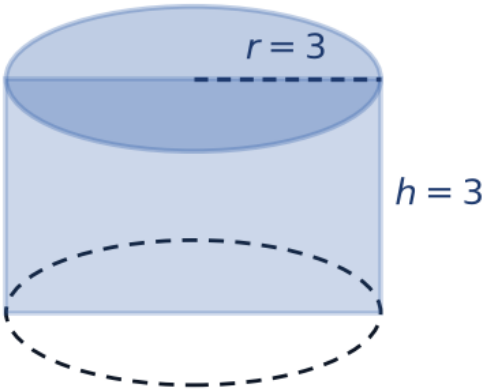
5 **[Foundational]** • **Target: The Disc Method Around Coordinate Axes**

The base of a solid is the region under $y = 1\sqrt{x}$ on $[0, 5]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

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6 [Foundational] • Target: The Washer Method Around Coordinate Axes

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 3]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.



7 [Foundational] • Target: Revolutions Around Non-Axis Horizontal Lines

The base of a solid is the region under $y = 1\sqrt{x}$ on $[0, 9]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Volumes with Known Cross-Sections

The base of a solid is the region under $y = 4\sqrt{x}$ on $[0, 10]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

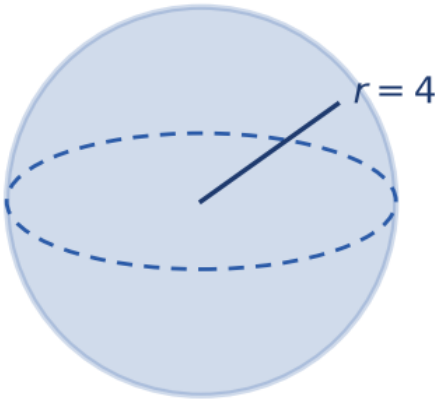
9 [Intermediate] • Target: The Disc Method Around Coordinate Axes

The base of a solid is the region under $y = 4\sqrt{x}$ on $[0, 4]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

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10 [Intermediate] • Target: The Washer Method Around Coordinate Axes

The base of a solid is the region under $y = 4\sqrt{x}$ on $[0, 8]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.



11 [Intermediate] • Target: Revolutions Around Non-Axis Horizontal Lines

The base of a solid is the region under $y = 4\sqrt{x}$ on $[0, 9]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

12 [Intermediate] • Target: Volumes with Known Cross-Sections

The base of a solid is the region under $y = 4\sqrt{x}$ on $[0, 7]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

13 [Intermediate] • Target: The Disc Method Around Coordinate Axes

The base of a solid is the region under $y = 2\sqrt{x}$ on $[0, 11]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

14 [Intermediate] • Target: The Washer Method Around Coordinate Axes

The base of a solid is the region under $y = 2\sqrt{x}$ on $[0, 10]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

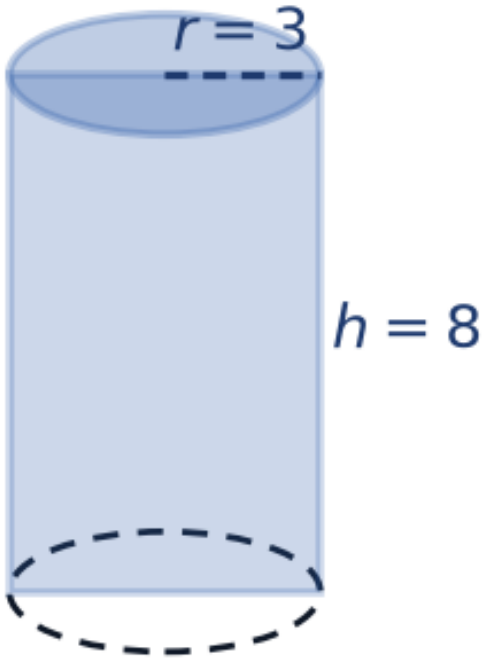
15 [Challenging] • Target: Revolutions Around Non-Axis Horizontal Lines • Cross-Lesson: 1.01 Limits Graphically and Numerically

The base of a solid is the region under $y = 4\sqrt{x}$ on $[0, 3]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

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16 [Challenging] • Target: Volumes with Known Cross-Sections • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 8]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.



17 [Challenging] • Target: The Disc Method Around Coordinate Axes • Cross-Lesson: 2.01 The Definition of the Derivative

The base of a solid is the region under $y = 2\sqrt{x}$ on $[0, 8]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

18 [Challenging] • Target: The Washer Method Around Coordinate Axes • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 6]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

19 [Challenging] • Target: Revolutions Around Non-Axis Horizontal Lines • Cross-Lesson: 3.01 The Chain Rule

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 4]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

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20 [Challenging] • Target: Volumes with Known Cross-Sections • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

The base of a solid is the region under $y = 2\sqrt{x}$ on $[0, 4]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Disc Method Around Coordinate Axes

Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ about the x-axis.

a. 8π

b. 16π

c. 4π

d. $\frac{16\pi}{3}$

22 [Foundational] • Target: The Disc Method Around Coordinate Axes

Find the volume when the region bounded by $y = x^2, x = 0,$ and $y = 4$ is revolved about the y-axis.

a. 8π

b. 16π

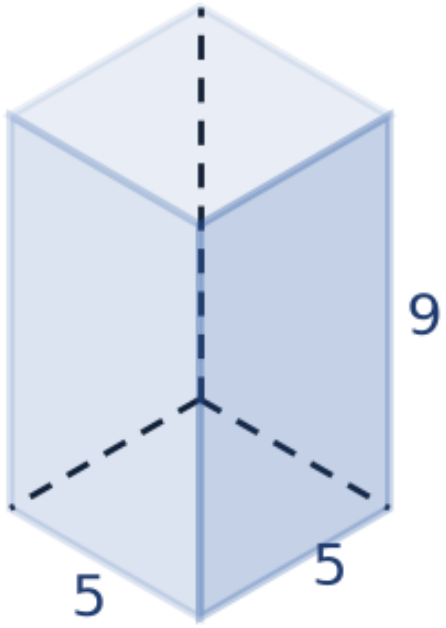
c. 4π

d. $\frac{32\pi}{5}$

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23 [Foundational] • Target: The Disc Method Around Coordinate Axes

What is the cross-sectional shape of a solid formed by the disc method?



a. A solid circular disk of area πR^2

b. A circular ring (washer)

c. A rectangle

d. A cylindrical shell

24 [Foundational] • Target: The Washer Method Around Coordinate Axes

The region enclosed by $y = x$ and $y = x^2$ is revolved about the x-axis. What is the resulting volume?

a. $\frac{2\pi}{15}$

b. $\frac{\pi}{6}$

c. $\frac{\pi}{15}$

d. $\frac{4\pi}{15}$

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25 [Foundational] • Target: The Washer Method Around Coordinate Axes

Write the integral for the volume when the region between $y = 2x$ and $y = x^2$ is revolved about the x-axis.

a. $\pi\int_0^2 (4x^2 - x^4)dx$

b. $\pi\int_0^2 (2x - x^2)^2 dx$

c. $\pi\int_0^2 (x^4 - 4x^2)dx$

d. $2\pi\int_0^2 (2x - x^2)dx$

26 [Foundational] • Target: The Washer Method Around Coordinate Axes

Why is $\pi\int (R - r)^2 dx$ INCORRECT for the washer method?

a. The area of a washer is $\pi R^2 - \pi r^2$, which is NOT equal to $\pi(R - r)^2$

b. It omits the factor of 2

c. It is only valid for shells

d. It produces negative volume

27 [Foundational] • Target: Revolutions Around Non-Axis Horizontal Lines

The region bounded by $y = x^2$ and $y = 4$ is revolved about the line $y = 4$. What is the outer radius $R(x)$?

a. $R(x) = 4 - x^2$

b. $R(x) = x^2$

c. $R(x) = 4$

d. $R(x) = x^2 - 4$

28 [Foundational] • Target: Revolutions Around Non-Axis Horizontal Lines

Find the volume in the above problem revolved about $y = 4$.

a. $\frac{256\pi}{15}$

b. $\frac{128\pi}{15}$

c. $\frac{64\pi}{5}$

d. 16π

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Revolutions Around Non-Axis Horizontal Lines**

The region between $y = \sqrt{x}$, $y = 0$, and $x = 4$ is revolved about the line $y = -1$. What are the washer radii?

a. $R(x) = \sqrt{x} + 1$ and $r(x) = 1$

b. $R(x) = \sqrt{x} - 1$ and $r(x) = -1$

c. $R(x) = \sqrt{x}$ and $r(x) = 1$

d. $R(x) = 4$ and $r(x) = \sqrt{x}$

30 **[Intermediate]** • **Target: Volumes with Known Cross-Sections**

The base of a solid is the region enclosed by $y = 1 - x^2$ and the x-axis. Cross-sections perpendicular to the x-axis are squares. What is the volume?

a. $\frac{16}{15}$

b. $\frac{8}{15}$

c. $\frac{4}{3}$

d. $\frac{32}{15}$

31 **[Intermediate]** • **Target: Volumes with Known Cross-Sections**

If cross-sections perpendicular to the x-axis are semicircles with diameter on the base, what is the cross-sectional area formula?

a. $A(x) = \frac{\pi}{8}[s(x)]^2$

b. $A(x) = \frac{\pi}{2}[s(x)]^2$

c. $A(x) = \frac{\pi}{4}[s(x)]^2$

d. $A(x) = \pi[s(x)]^2$

32 **[Intermediate]** • **Target: Volumes with Known Cross-Sections**

If cross-sections are equilateral triangles with side length $s(x)$, what is the cross-sectional area formula?

a. $A(x) = \frac{\sqrt{3}}{4}[s(x)]^2$

b. $A(x) = \frac{1}{2}[s(x)]^2$

c. $A(x) = \frac{\sqrt{3}}{2}[s(x)]^2$

d. $A(x) = \frac{1}{4}[s(x)]^2$

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33 [Intermediate] • Target: The Disc Method Around Coordinate Axes

The base of a solid is the region under $y = 1\sqrt{x}$ on $[0, 8]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

a. 64

b. 30

c. 32

d. 34

34 [Intermediate] • Target: The Washer Method Around Coordinate Axes

The base of a solid is the region under $y = 2\sqrt{x}$ on $[0, 7]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

a. 196

b. 100

c. 96

d. 98

35 [Intermediate] • Target: Revolutions Around Non-Axis Horizontal Lines

The base of a solid is the region under $y = 1\sqrt{x}$ on $[0, 4]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

a. 6

b. 8

c. 10

d. 16

36 [Intermediate] • Target: Volumes with Known Cross-Sections

The region between $y = 8$ and $y = x^2$ on $[0, 2]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{608}{5}$

b. $\frac{304}{5}\pi$

c. $\frac{1216}{5}\pi$

d. $\frac{608}{5}\pi$

37 [Intermediate] • Target: The Disc Method Around Coordinate Axes

The region between $y = 9$ and $y = x^2$ on $[0, 2]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{778}{5}$

b. $\frac{389}{5}\pi$

c. $\frac{1556}{5}\pi$

d. $\frac{778}{5}\pi$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Washer Method Around Coordinate Axes • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

The region between $y = 9$ and $y = x^2$ on $[0, 1]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{202}{5}\pi$

b. $\frac{808}{5}\pi$

c. $\frac{404}{5}\pi$

d. $\frac{404}{5}$

39 [Challenging] • Target: Revolutions Around Non-Axis Horizontal Lines • Cross-Lesson: 3.01 The Chain Rule

The region between $y = 6$ and $y = x^2$ on $[0, 1]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{179}{10}\pi$

b. $\frac{179}{5}$

c. $\frac{358}{5}\pi$

d. $\frac{179}{5}\pi$

40 [Challenging] • Target: Volumes with Known Cross-Sections • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

The region between $y = 2$ and $y = x^2$ on $[0, 1]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{19}{5}$

b. $\frac{19}{10}\pi$

c. $\frac{19}{5}\pi$

d. $\frac{38}{5}\pi$

41 [Challenging] • Target: The Disc Method Around Coordinate Axes • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

The region between $y = 3$ and $y = x^2$ on $[0, 1]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{44}{5}$

b. $\frac{22}{5}\pi$

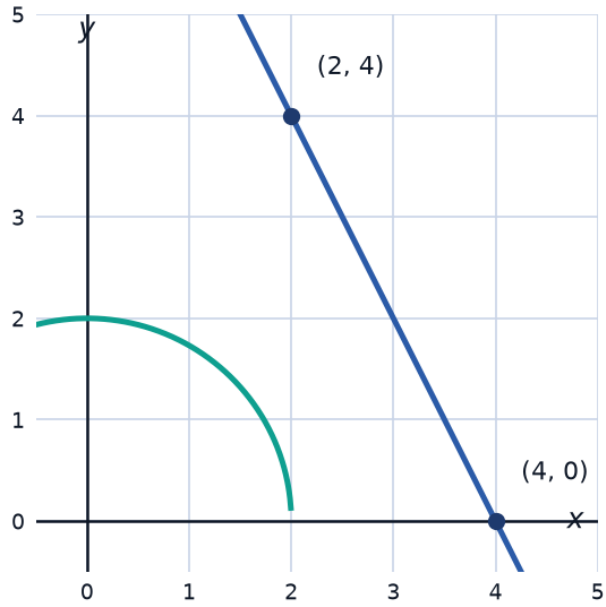
c. $\frac{88}{5}\pi$

d. $\frac{44}{5}\pi$

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42 [Challenging] • Target: The Washer Method Around Coordinate Axes • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Consider the function $f(x)$ whose graph is shown below. The graph consists of a line segment and a quarter circle. The line segment connects the points $(2, 4)$ and $(4, 0)$. The quarter circle is in the second quadrant and is centered at $(0, 0)$ with radius 2. Let $A(x)$ be the area of the region bounded by the graph of $f(t)$, the t-axis, and the lines $t = 0$ and $t = x$. (a) Find $A(3)$.



a. 4

b. 6

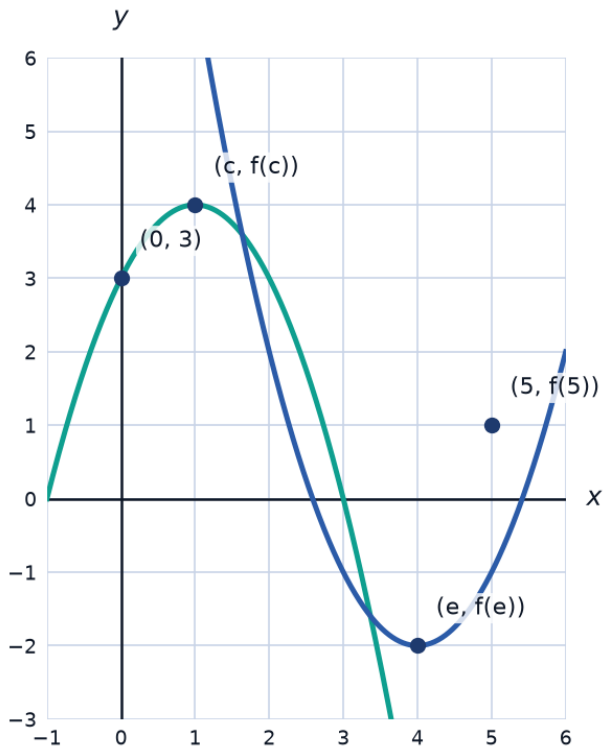
c. 5

d. 7

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43 **[Challenging]** • **Target: Revolutions Around Non-Axis Horizontal Lines** • **Cross-Lesson: 5.01 Extrema and the Mean Value Theorem**

The graph of a differentiable function f is shown in the figure above. The function f has a local maximum at $x = c$ and a local minimum at $x = e$. The point $(c, f(c))$ is approximately $(1, 4)$ and the point $(e, f(e))$ is approximately $(4, -2)$. The shaded region, bounded by the graph of f, the x-axis, and the lines $x = 0$ and $x = 5$, has area 12. $*If*\int_{0}^{c} f(x) \, dx = k, *whatisthevalueof*\int_{c}^{5} f(x) \, dx*intermsof*k$?



a. 12 + k

b. 12 - k

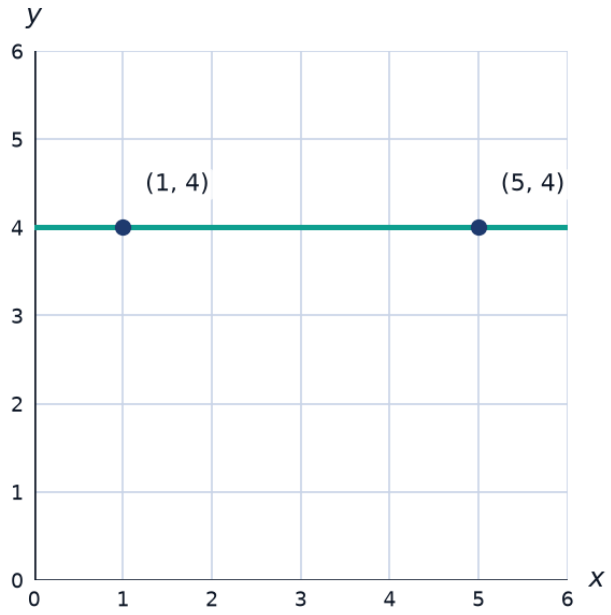
c. k - 12

d. k + 2

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44 [Challenging] • Target: Volumes with Known Cross-Sections • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

The graph of a function $f'(x)$ is shown below. It is given that $f(x)$ is a function such that $f(1) = 5$. The area between the curve $y = f'(x)$, the x-axis, and the vertical lines $x = 1$ and $x = 5$ is represented by the shaded region. If the area of the shaded region is 16 square units, determine the value of $f(5)$.



a. 17

b. 21

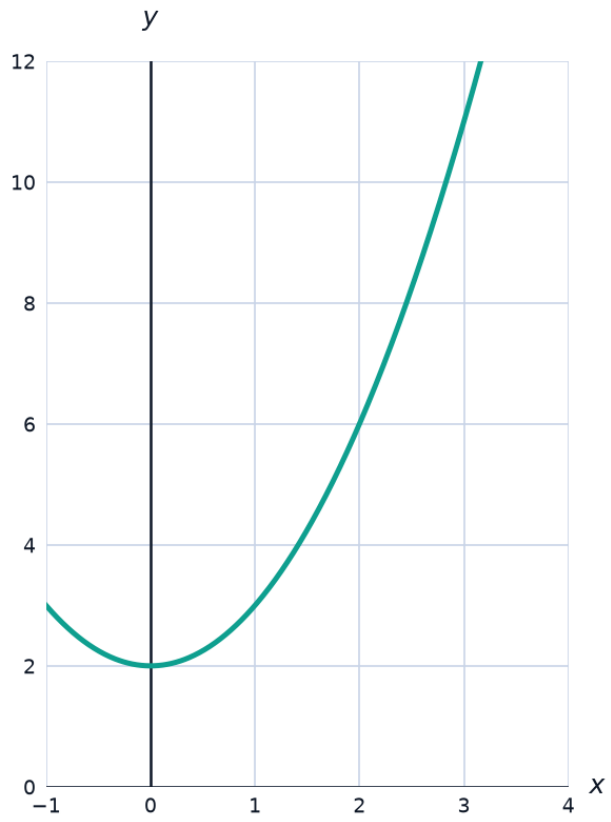
c. 9

d. 11

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45 **[Challenging]** • **Target: The Disc Method Around Coordinate Axes** • **Cross-Lesson: 6.01 Antiderivatives and Riemann Sums**

Consider the region bounded by the graph of $f(x) = x^2 + 2$, the x-axis, and the vertical lines $x = 0$ and $x = 3$. Which of the following represents the area of this region?



a. $\int_0^3 (x^2 + 2)\,dx$

b. $\int_0^3 (x^2 - 2)\,dx$

c. $\int_0^3 x^2\,dx$

d. $\int_0^3 2\,dx$

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